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DIFFERENCE

Structures and Materials Lab

ABOUT THE SPACE:

The Structures and Materials (SMASH) lab is taken in the third year of the civil engineering (CIVE) and construction engineering (CONE) curriculums. In this lab, students test a variety of materials used in construction. Steel, concrete, and wood are the three most common materials used worldwide. Changes in material shape, placement, treatment, and installation can drastically change how they behave and fail. This is why material science is so important in the fields of civil and construction engineering.

Using the beam buster in the back right corner, students will load steel and wood beams in tension or compression, to examine how and why they fail.

Torsion testers are used to twist steel samples, and a hydraulic press is used to crush concrete samples. Testing materials to failure is imperative in understanding the full strength, ductility, and integrity of different materials. This ensures that they are used well within their safety limits.

The canoe suspended from the ceiling was top ranked in one of the intercollegiate competitions for civil engineers. The canoe is made up entirely of concrete, with a unique foam bead aggregate, or mix-in, that makes it buoyant.



Concrete canoe: Eagle, c. 2002

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